

Abstract:

Gas sensor array (GSA) data usually has high-dimensional features and a small sample size. When a classifier is directly used for GSA data classification, it is prone to overfitting and has a high time cost. The traditional solution is to perform feature dimensionality reduction before classification. However, selecting a suitable dimensionality reduction method is time-consuming and laborious, and some features useful for classification may be lost after dimensionality reduction, especially for the weak sensor response data to low-concentration gases. In this article, we proposed a feature ensemble-based extreme learning machine framework (FE-ELM) for GSA data classification. In FE-ELM, downsampling is first performed on the time series of each sensor, and then the downsampled features of different sensors are combined to obtain fused feature subsets. Next, a base ELM is trained independently on each fused feature subset with all training samples by solving the least-squares problem. The final FE-ELM predictions for input samples are obtained by voting the prediction results of all base ELMs. Compared with traditional methods, the proposed method solves the overfitting problem and can be directly used for GSA data classification without prior feature dimension reduction. Furthermore, the ensemble of all base classifiers with little loss of original features enables the proposed FE-ELM to have a more efficient and robust classification performance. Experimental results on data from both homemade GSA under low-concentration gases (ppb) and publicly available confirm that the proposed FE-ELM exceeds traditional methods and extends the detection limit of the sensor array.

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SECTION I.

Introduction

Gas sensor array (GSA) is an important device for gas detection, which converts odor information into electrical signals, and enables subsequent pattern recognition to be performed after signal preprocessing [1], [2]. Each sample of GSA data consists of multivariate time series, which usually has high dimensionality (at least thousands) [3]. At the same time, it is usually of small sample size (tens to hundreds) due to a lack of standard procedures for sample collection and high cost [4]. Classic classifiers for GSA data classification include k-nearest neighbors (KNN), support vector machines (SVM), random forest (RF), logistic regression (LR), extreme learning machine (ELM), multilayer perceptron (MLP), and so on [5]. When they are used directly on original GSA data, in addition to high computational cost, the model overfitting caused by high-dimensional features and small sample size will deteriorate their classification performance.

Traditionally, to solve the above overfitting problem, the classification task of GSA data is usually performed after feature dimensionality reduction [6]. In [7] and [8], feature extraction methods for GSA data are reviewed, such as extraction from time-series curves, curve-fitting parameters, transformation domain, phase space, adaptive learning, and so on. Feature selection methods are also often used to remove redundant features [9], [10], [11]. A suitable feature dimensionality reduction method can alleviate the overfitting of the subsequent classifier and improve its classification performance. However, as far as we know, there are no rules to follow in choosing these methods [12]. In particular, it is more difficult to extract representative features for GSA data under low-concentration gases due to the weak response amplitude of the sensors. To obtain the best recognition results, many methods must be tested to find the best one for a specific application, which often requires a lot of time and effort. Meanwhile, after feature dimensionality reduction, some features that are beneficial for classification may be lost. In addition, deep learning has been widely used in sensor data classification tasks due to its automatic feature learning [13], [14], [15], [16], [17], [18], [19]. In [20], an improved LeNet-5 network was used for GSA data classification, and it achieved better classification accuracy than SVM, MLP, and probabilistic neural network. In [21], a deep-learning framework consisting of a convolutional LSTM layer and a regression layer was used for odor descriptor rating prediction and achieved state-of-the-art performance than using the temporal or spatial model alone. However, deep learning usually requires a large number of

samples to train the parameters of the model, which may not be powerful when used for small sample size data.

Recently, ensemble learning has been successful in many scenarios due to its ability to turn weak learners into strong learners [22], [23], [24], [25]. In [24], an optimized two-layer Adaboost model was proposed to improve the accuracy and reliability of multiclass identification of herbal medicines, and the integrated model achieved the best accuracy compared to the single classifiers. In [25], an artificial neural network (ANN) model was independently trained on each sensing unit to identify volatile organic compounds (VOCs), and then different ANNs were integrated into one model, which showed a great improvement in classification accuracy compared to the best individual classifier. Unlike traditional methods that train a single learner, ensemble learning methods train multiple base learners and then integrate them to improve overall learning performance. Inspired by the idea of the ensemble, we propose a feature ensemble-based ELM (FE-ELM) to solve the GSA data classification problems. The reason for employing ELM as the base classifier is that it has the advantages of the global optimal solution and low time cost [26]. In FE-ELM, the time series of each sensor is first downsampled and then the downsampled features from different sensors are fused to obtain fused feature subsets. Next, a base ELM classifier is trained separately on each fused feature subset. And the final prediction of FE-ELM for a testing sample is obtained by voting on the prediction results of all base classifiers. Compared with traditional methods, the proposed method has the following advantages.

1. Compared with classical single classifiers, FE-ELM trains multiple base classifiers on feature subsets and then integrates them into a strong classifier, which overcomes the overfitting problem and has a more robust classification ability.
2. Compared with the methods of feature dimensionality reduction first and then classification, FE-ELM can be directly used for GSA data classification without losing the original information, so it has the advantage of being simple and effective.
3. Compared with traditional ensemble learning methods that train base classifiers on sample subsets, FE-ELM trains base classifiers on fused feature subsets, which overcomes the overfitting problem of the model.

The proposed method is evaluated on a homemade GSA under low concentration gases dataset and a publicly available GSA dataset.

Experimental results show that the proposed method surpasses traditional methods with more efficient and robust classification performance. The rest of this article is organized as follows. In Section II, the data collection process is described. In Section III, the proposed method and its evaluation criteria are presented. In Section IV, experimental results are reported. Finally, conclusions are offered in Section V.

SECTION II.

Data Collection and Preprocessing

Most of the GSA data is collected in response to gases at ppm concentration levels. The detection results of GSA for lower gas concentrations are not yet well understood. In this section, we collected data on the response of GSA to low-concentration gases with ppb level.

A. Experimental Platform

The experimental platform includes a gas-sensitive element test system (WS-30B) produced by Winsen (China) and a GSA. Fig. 1(a) gives the photograph of the experimental platform. The test system mainly contains a computer, a data acquisition module, an air box, and so on. It can provide the heating power V_h and loop power V_c required for gas sensor work and support testing of up to 32 sensors simultaneously. When a gas is passed into the box, the voltages V_{out} across the load resistor R_1 in series with the sensor are acquired by the AD acquisition card and then transmitted to the upper computer software for graphical presentation. These voltage signals reflect the response characteristics of the sensor to the gas. Fig. 1(b) shows the schematic for only one sensor. The sensor array consists of ten different metal oxide semiconductor sensors, which have broad-spectrum response characteristics and can detect a variety of VOCs, such as alcohols, ketones, benzenes, esters, alkanes, and so on [27]. Based on the characteristics of safety, stability, universality, and easy availability, we selected six representative gases: ethanol, acetone, toluene, ethyl acetate, isopropanol, and normal hexane. The algorithm constructed based on the response data of the sensor array to these six gases has certain universality and practicability. Detailed information about the gas sensors is presented in Table I.

TABLE I Details of the Gas Sensors

Fig. 1.

Photograph and schematic of the experimental platform. (a) Photograph of the experimental platform. (b) Schematic of the sensor experiment.

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B. Prepare Standard Gas

We first prepare the standard gases with 30-ppm concentration. Six 20-L Teflon bags are filled with pure air and then injected with appropriate amounts of ethanol, acetone, toluene, ethyl acetate, isopropanol, and normal hexane liquid samples by using a microinjector, respectively. After the liquids are completely evaporated, it will produce six gases with 30-ppm concentration. The injected amount V_x in units microliter (μL) can be calculated by

$$V_x = V \times C \times M \times V_m \times d \times p \times 10^{-3} \quad (1)$$

[View Source](#) [?] where d is the purity of liquid, p is the density of the liquid in units g/ml, V_m is the molar volume of gases in 24.5L/mol at normal temperature and pressure, M is the relative molecular mass of the gases in units g/mol, $C=30$ is the concentration of the gases in units ppm, and $V=20$ is the volume of the bags in units liter (L). Table II presents the specifications and the number of chemicals used in the configuration of the standard gases. Then, 30, 60, and 120 ml of standard gas with 30-ppm concentration will be injected into the box with a volume of 18 L for further dilution, and it can be easily calculated that the gas concentration in the box will be 50, 100, and 200 ppb, respectively.

TABLE II Specifications and Amount of Chemicals Used in the Configuration of the Standard Gases

C. Data Collection

We now start data collection. Turn on the power switch of the gas-sensitive element test system and set the experimental parameters as follows: 1) set the heating voltage and loop voltage to 5 V; 2) set the baseline, adsorption, and desorption time to 5, 10, and 15 min, respectively; and 3) set the sampling frequency to 1 Hz. The data collection process will begin after the system runs stably.

The data collection process consists of three steps. In the first step, 5-min baseline response data of the sensor array to background air in the box is collected. In the second step, 30-, 60-, and 120-ml standard gas is injected into the box by using a syringe and then the response data of the sensor array to the target gas is collected for 10 min. In the third step, open the box to clean the sensor array to release the adsorbed gas.

Following the steps above, we collect data on the response of the sensor array to six classes of gases with three concentration gradients for each class. And the collection process is repeated five times for each concentration of each type of gas. Finally, we collect a total of 90 samples from six classes. In each experiment, we collect ten time series of length 900, that is, each sample contains 9000 features.

D. Data Preprocessing

Since the original response signals usually contain high-frequency noise, an FIR low-pass filter with order 100 and a cutoff frequency of 0.01 is used to remove noise. Taking the sensor TGS2603 as an example, its filtered response curves to ethanol gas with three concentrations are shown in Fig. 2, where the first subplot is the original response curve and the last three subplots are the downsampled curves with sampling periods of 32, 64, and 128, respectively.

Fig. 2.

Sensor responses for ethanol gas with three concentrations. (a) Original responses. (b)–(d) Downsampled responses with sampling periods of 32, 64, and 128, respectively.

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SECTION III.

Proposed Method

In this section, we first briefly introduce ensemble learning and standard ELM and then propose FE-ELM to solve the GSA data classification problem caused by high-dimensional features and a small sample size. The evaluation criteria for the proposed method on small sample-size datasets are also presented.

A. Ensemble Learning

Recently, ensemble learning methods have received a lot of attention, not only achieving outstanding results in competitions, but also being successfully applied to almost all machine learning application scenarios [28]. The reason why ensemble learning methods are so successful is that they can turn weak learners into strong learners, and weak learners are readily available in practice. Two pioneering works not only show experimentally that an ensemble of classifiers often yields more accurate predictions than individual classifiers, but also theoretically constructively demonstrate that weak learners can be promoted to strong ones [29], [30].

The ensemble learning methods can be divided into two paradigms: boosting and bagging. In boosting, a series of base learners are trained serially, and the samples misjudged by the previous base learners will receive more attention in the follow-up and then these learners are combined to obtain a strong learner. The bagging trains base learners in parallel on different sample subsets obtained by a bootstrap sampling method and then aggregates these base learners by averaging or voting. The boosting solves the underfitting of the model by reducing the training residuals based on the correlation between base learners, and bagging improves the generalization performance of the model by reducing the prediction variance based on the independence between base learners. Let f be the true discriminant function and $h(x)$ be a base learner trained on a sample subset D obtained by bootstrap sampling. The learner after bagging the ensemble is

$$H(x) = E_D[h(x)]. \quad (2)$$

[View Source](#)  According to the inequality $(E[X])^2 \leq E[X^2]$, we can get

$$(f(x) - H(x))^2 \leq E_D[(f(x) - h(x))^2]. \quad (3)$$

[View Source](#)  It is not difficult to find that the ensemble learner $H(x)$ has a smaller mean squared error than a single learner $h(x)$. However, the base classifiers of both boosting and bagging methods are trained on different sample sets, and they will suffer from the overfitting problem on GSA data.

B. Extreme Learning Machine

ELM was proposed in [26] for training a single hidden layer feedforward neural network (SLFN). A typical SLFN contains an input layer, a hidden layer, and an output layer. In ELM, the hidden layer nodes are randomly initialized and fixed without iterative adjustment, and the only parameters to be learned are the connection weights between the hidden layer and the output layer. Even with randomly generated hidden layer nodes, ELM maintains the universal approximation capability of SLFN [31]. Compared to traditional gradient-based methods, ELM can get the global optimal solution

with low computational cost by solving the least-squares problem. Benefiting from these advantages, ELM is suitable as a base classifier for ensemble learning.

C. Feature Ensemble-Based ELM

Inspired by bagging, we propose an FE-ELM method to solve the problem encountered in the GSA data classification task, and its architecture is presented in Fig. 3. The FE-ELM consists of three steps. First, downsampling the time series of each sensor in the same period and initial point, and then combining the downsampled features of different sensors to obtain a fused feature subset. Different fused subsets can be obtained by moving the initial point of the downsampling. Next, a base ELM classifier is trained separately on each fused feature subset with all training samples by solving the least squares problem. And the final prediction of FE-ELM for a testing sample is obtained by voting on the results of all base ELMs.

Fig. 3.
Structure of the proposed FE-ELM.

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Suppose $X=(S_1, S_2, \dots, S_m) \in \mathbb{R}^{n \times (dm)}$ represents the original GSA data and $y \in \mathbb{R}^{n \times 1}$ represents the corresponding label vector, where n is the sample size, m is the number of sensors (or variables), and d is the length of the time-series signal for each sensor. The class label of the i th sample x_i is assumed to be $y_i \in \{1, 2, \dots, c\}$, where c is the number of classes. In multiclass classification tasks, class label vector y is usually transformed into matrix $Y \in \mathbb{R}^{n \times c}$ consisting of “0/1” element by one-hot coding.

We first introduce the training process of the FE-ELM. For the i th training sample x_i , it consists of a multivariate time-series signal, that is,

$$x_i = ((s_{1i})^T, (s_{2i})^T, \dots, (s_{mi})^T)^T \quad (4)$$

[View Source](#)  where s_{ji} is the i th row of S_j . Perform downsampling for each time series with the same sampling period τ and initial position $t(1 \leq t \leq \tau)$ and then combine the downsampled features of different sensors to get a fused feature subset

$$X_{ti} = (S_{1i,t}, S_{1i,t+\tau}, \dots, S_{1i,t+(u-1)\tau}, S_{2i,t}, S_{2i,t+\tau}, \dots, S_{2i,t+(u-1)\tau}, \dots, S_{mi,t}, S_{mi,t+\tau}, \dots, S_{mi,t+(u-1)\tau})^T \quad (5)$$

[View Source](#)  where $s_{ji,t}$ is the t th element of s_{ji} , and $u = \lfloor d/\tau \rfloor$ is the number of downsampling points for each time series, $\lfloor \cdot \rfloor$ represents rounding down. The reason for this is that downsampling can solve the overfitting problem of the model, and sensor feature fusion is a necessary technology for GSA data classification. By moving the initial position of the downsampling for each time series, τ different fused feature subsets $\{X_{ti}\}_{t=1}^\tau$ are obtained. In fact, the union of these τ subsets is almost equal to the original dataset, that is,

$$X_i \approx ((X_{1i})^T, (X_{2i})^T, \dots, (X_{\tau i})^T)^T. \quad (6)$$

[View Source](#)  For all training samples, we have similarly $X \approx (X_1, X_2, \dots, X_\tau)$, where the i th row of $X_t \in \mathbb{R}_{n \times (\mu m)}$ is X_{ti} . Next, a base ELM is trained separately on each fused subset. Initializing the connection weight $W_t \in \mathbb{R}_{(\mu m) \times h}$ and bias $b_t \in \mathbb{R}_{h \times 1}$ between the input layer and hidden layer in the t th base ELM and fixing them without adjustment, where h is the hidden layer size of the base ELM. X_t is input to the t th base ELM network and the output of the hidden layer can be computed by

$$H_t = g(X_t W_t + 1(b_t)^T) \quad (7)$$

[View Source](#)  where $g(\cdot)$ is the sigmoid activation function and 1 is a vector of length n with all elements one. Then the output layer can be calculated by

$$Y^{\wedge}_t = H_t \cdot \beta_t = g(X_t W_t + 1(b_t)^T) \cdot \beta_t \quad (8)$$

[View Source](#)  where $\beta_t \in \mathbb{R}_{h \times c}$ is the connection weight matrix between the hidden layer and the output layer in the t th base ELM, which needs to be learned. The sum of prediction errors of the t th base ELM for all training samples is

$$e_t = \|Y - Y^{\wedge}_t\|_{2F} = \|Y - H_t \beta_t\|_{2F} \quad (9)$$

[View Source](#)  where $\|\cdot\|_F$ is the Frobenius norm. The base ELM obtains optimal β_t by minimizing the training error

$$\min_{\beta_t} \|Y - H_t \beta_t\|_{2F}. \quad (10)$$

[View Source](#)  According to the least-squares regression theory, the minimum norm solution of the objective function is

$$\beta^{\wedge}_t = (H_t)^+ Y \quad (11)$$

[View Source](#)  where H^+ is the Moore-Penrose generalized inverse of matrix H , when H is column full rank and $H^+ = (H^T H)^{-1} H^T$, or when H is row

full rank and $H_+ = H_T(H_T H_T^T)^{-1}$. After repeating the above process on τ fused subsets, the trained FE-ELM network is obtained. Note that the ensemble scale of FE-ELM is equal to the sampling period τ .

We now introduce the testing process of the FE-ELM. A test sample is first divided into τ fused subsets according to (6), $x = ((x_1)_T, (x_2)_T, \dots, (x_\tau)_T)^T$, and then they are, respectively, input into τ trained base ELMs in order. The output of the t th base ELM can be computed by

$$y^t = (\beta^t)^T \cdot g((W_t)^T x_t + b_t) \quad (12)$$

[View Source](#) [?] and the corresponding predicted label can be obtained by

$$y^t = \arg \max_{1 \leq i \leq c} y^{ti} \quad (13)$$

[View Source](#) [?] where y^{ti} is the i th element of y^t . In this way, we will get τ predicted labels for the test sample. Finally, by the relative majority voting method, the predicted label of FE-ELM for the test sample is obtained

$$y = \arg \max_{1 \leq i \leq c} \sum_{t=1}^{\tau} I_i(y^t) \quad (14)$$

[View Source](#) [?] where $I_i(x)$ is an indicator function, $I_i(x) = 1$ if $x = i$, and $I_i(x) = 0$ if $x \neq i$.

As seen in (5), each base classifier of FE-ELM is trained on a fused feature subset obtained by downsampling and fusing the original time series, which not only avoids the overfitting problem caused by the high-dimensional features, but also exploits the power of feature fusion. From (6) and (14), it can be seen that FE-ELM makes full use of almost all the original features by integrating all the base classifiers, so it will have efficient and robust classification performance.

The time cost of FE-ELM is mainly focused on the training of τ base ELM classifiers. For the t th base ELM, its time cost is concentrated on the computation of the hidden layer features and the solution of the optimal parameter between the hidden layer and the output layer, which are computed by (7) and (11), respectively. The time complexity of the two formulas is $O(umnh)$ and $O(2nhv + v^3 + hnc)$, respectively, and the two can be added to obtain the time complexity of each base ELM, $O(umnh + 2nhv + v^3 + hnc)$, where $v = \min(n, h)$. So, the time complexity of the FE-ELM integrated by τ base ELMs is $O(dumnh + \tau(2nhv + v^3 + hnc))$. Algorithm 1 summarizes the workflow of the proposed FE-ELM.

Algorithm 1 FE-ELM for GSA Data Classification

Input: training set $\{X, Y\}$, testing sample x , sampling period τ and hidden layer size h .

Output: predicted label $y^\wedge$ of the FE-ELM for testing sample x

1:

Training process

2:

while $1 \leq t \leq \tau$ **do**

3:

obtain the input X_t of the t th base ELM by downsampling X according to (5)

4:

initialize the connection weight W_t and bias b_t between the input layer and the hidden layer

5:

calculate the output of the hidden layer of the t th base ELM according to (7)

$$H_t = g(X_t W_t + 1(b_t) \tau)$$

6:

calculate the weight between the hidden layer and the output layer according to (11)

$$\beta^\wedge_t = (H_t)^T Y$$

7:

end while

8:

Testing process

9:

while $1 \leq t \leq \tau$ **do**

10:

obtain the input x_t of the t th base ELM by downsampling x according to (5)

11:

calculate the output of the t th base ELM for x_t according to (12)

$$y^\wedge_t = (\beta^\wedge_t)^T g((W_t)^T x_t + b_t)$$

12:

get the predicted label $y^\wedge_t$ of the t th base ELM for x_t according to (13)

$$y^\wedge_t = \operatorname{argmax}_{1 \leq i \leq c} y^\wedge_{ti}$$

13:

end while

14:

$\hat{y}$ is obtained by voting on $\{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_\tau\}$ according to (14)

D. Evaluation Criterion

In classification tasks, the dataset is usually divided into a training set and a test set. A classifier is tuned on the training set and then tested on the test set. The test accuracy reflects the discriminatory ability of the classifier for new samples and can be calculated by

$$\text{acc} = \frac{1}{n} \sum_{i=1}^n I(\hat{y}_i = y_i) \quad (15)$$

[View Source](#) [?] where I is an indicator function, n is the number of test samples, and y_i and $\hat{y}_i$ are the true and predicted label, respectively. Due to the small sample size of the GSA data, testing on only one test set is not sufficient. Therefore, we use k -fold cross-validation to verify the performance of the proposed method. The dataset is first divided into k mutually exclusive sample subsets of the same size $\{D_1, D_2, \dots, D_k\}$ and then execute the training-testing process k times. In the i th time, D_i is used as the test set and all other subsets are used as the training set. Also, to reduce the effect of randomness in data partitioning, we repeatedly perform r times different random partitions and report the mean and standard deviation of $r \cdot k$ test results. They can be calculated as follows:

$$\bar{d} = \frac{1}{r \cdot k} \sum_{i=1}^{r \cdot k} d_i, \quad s = \sqrt{\frac{1}{r \cdot k - 1} \sum_{i=1}^{r \cdot k} (d_i - \bar{d})^2} \quad (16)$$

[View Source](#) [?] where d_i is the i th test result, and $\bar{d}$ and s are the mean and standard deviation of all test results, respectively. Assuming that $r \cdot k$ results are independently and identically distributed with expectation μ , the average accuracy $\bar{d}$ approximately obeys a normal distribution according to the central limit theorem. Based on the student t -distribution

$$t = \frac{\bar{d} - \mu}{s / \sqrt{r \cdot k}} \sim t(r \cdot k - 1) \quad (18)$$

[View Source](#) [?] 95% confidence interval of expectation μ is also calculated by $\bar{d} \pm t_{0.975}(r \cdot k - 1) \cdot s / (r \cdot k)^{1/2}$, where $t_{0.975}(r \cdot k - 1)$ is the 0.975 quantile of the t -distribution with degree of freedom $r \cdot k - 1$, and this interval can cover the true accuracy μ with 95% probability.

SECTION IV.

Experimental Results

In this section, we compare the proposed FE-ELM with other classical classifiers based on the original features and dimensionality-reduced features of the two GSA datasets. The impact of the parameters on the proposed method is also presented.

A. Datasets and Visualizations

We evaluate the proposed method not only on the homemade dataset of GSA responses to low-concentration gases (GSA-LC), but also on a publicly available GSA under the flow modulation dataset (GSA-FM) from the UCI machine learning repository. The GSA-FM [3] was collected from an array of 16 metal-oxide gas sensors under gas flow modulation conditions. The sensors were exposed to different gaseous binary mixtures of acetone and ethanol. The dataset contains four categories (air, acetone, ethanol, and mixture) with a total of 58 samples and the sample size of each class is 8, 15, 15, and 20. Each sample included 16 time series (one time series per sensor) and each series contains 7481 points. Both datasets are of high-dimensional features and small sample sizes. Fig. 4 presents the dimensionality reduction results by PCA for the original features of the two datasets.

Fig. 4.

Dimensionality reduction by PCA for the original features of the two GSA datasets. (a) GSA-LC. (b) GSA-FM.

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B. Experimental Settings

In each experiment, the dataset is divided into a training set and a test set by fivefold cross-validation. For each classifier, 20 times fivefold cross-validation is performed and then the averaged classification results are presented. The optimal parameters for each classifier are selected by a grid search method, the average classification accuracy on validation sets is used as a criterion for the evaluation of the parameters, and the validation sets are obtained by performing a threefold cross-validation on the training set. In KNN, the candidate set for parameter k is $\{1,3,5,7\}$. In Adaboost, the candidate set for the number of base decision trees is $\{10,20,30,40,50,60\}$. In RF, the candidate set for the number of trees is $\{100,200,400,800,1600,3200\}$. In LR and MLP, the “lbfgs” optimizer is used to learn the optimal parameters. In SVM, the candidate sets for log value of the regularization parameter C and

kernel parameter G are both $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. In ELM, the candidate set for the hidden layer size is $\{200, 400, 600, 800, 1000\}$. In FE-ELM, we empirically set the sampling period to 64 and set the hidden layer size of each base ELM to 200 in all experiments. The evaluation of the effect of the parameters on the classification results will be presented subsequently. The feature subsets of GSA data are fed into the FE-ELM network, which is not standardized because there is no magnitude difference among the different feature dimensions. For the features obtained by feature extraction methods, there is a magnitude difference among them, and we standardize them by a Z-score method. All experiments are performed on a computer with Intel core i7-9700 CPU, Microsoft windows 10, and Python 3.7.

C. Evaluate the Impact of Parameters

We evaluate the impact of parameters on the classification accuracy of FE-ELM on two datasets. Fig. 5(a)–(c) presents the relationship between classification accuracy and the hidden layer size with different sampling periods. In each subplot, the horizontal axis represents the hidden layer size and the vertical axis represents the classification accuracy with error bars. It can be seen that FE-ELM is insensitive to the selection of hidden layer size within a certain parameter range, which is thanks to model integration.

Fig. 5.

Relationship between classification accuracy and parameter selection of FE-ELM on two datasets. (a)–(c) Sampling period is 32, 64, and 128, respectively. (d)–(f) Hidden layer size is 200, 600, and 1000, respectively.

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Fig. 5(d)–(f) presents the relationship between classification accuracy and the sampling period with different hidden layer sizes. In each subplot, the horizontal axis represents the sampling period and the vertical axis represents the classification accuracy with error bars. It can also be seen that the classification accuracies of FE-ELM increase first and then decreases with the increase of the sampling period. This is because the initial downsampling can alleviate the negative impact of high-dimensional features on each base classifier, thereby improving the overall classification performance, but when the sampling period is too large, each base classifier can only obtain a small number of useful features, which leads to poor classification performance of the ensemble model.

D. Comparison of Different Classifiers on Original GSA Data

We compare the proposed FE-ELM with classical classifiers including KNN, adaptive boosting (Adaboost), RF, LR, MLP, ELM, linear SVM (SVM-L), and kernel SVM (SVM-K) on the original GSA datasets.

Fig. 6 presents box plots of 100 test accuracies for each classifier, and it can be seen that the median, upper quartile, and lower quartile of the test results by FE-ELM are significantly higher than almost all other methods. We perform a one-tailed Student's t -test on FE-ELM and other classifiers. In each test, the alternative hypothesis is that the test accuracy of FE-ELM is higher than other classifiers, and the significance level is set to 0.05. Table III presents the results of the hypothesis testing, with 1 representing the acceptance of the alternative hypothesis and 0 representing the opposite. And all p-values are close to 0 except for LR and SVM-L on GSA-FM with p-values of 0.1681 and 0.2388. It can be seen that the proposed method outperforms the other classifiers in terms of statistical significance. Table IV presents the mean and 95% confidence interval of one hundred test accuracies, and the average time cost for one experiment. It can be seen that the proposed FE-ELM has better classification accuracy and lower time cost than other methods. FE-ELM achieves the superiority of classification because its base classifiers are trained on feature subsets and then integrated to become a strong classifier, which overcomes the overfitting problem and exploits the power of the model ensemble. It has the advantage of low time cost due to the small hidden layer size.

TABLE III Results of Student's t -Test on FE-ELM and Other Classifiers With Significance Level 0.05. All the p -Values are Close to 0 Except for LR and SVM-L on GSA-FM

TABLE IV Average Classification Results of Classifiers by 20 Times Fivefold Cross-Validation on Two Datasets

Fig. 6.

Box plots of classification results of the classifiers by 20 times fivefold cross-validation. (a) GSA-LC. (b) GSA-FM.

Fig. 7.

Confusion matrices of the classification accuracies of FE-ELM for each sample category on two datasets. (a) GSA-LC. (b) GSA-FM.

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In addition, Fig. 6 shows the confusion matrices of the classification results of the proposed FE-ELM on two datasets. The classification accuracies presented in the figures are the average results of 20 times fivefold cross-validation. In the dataset GSA-LC, each class has 15 samples, so there are 300 test samples for each class in 20 times of fivefold cross-validation, and the number of correctly predicted samples are 275, 273, 298, 293, 283, and 278, respectively. In the dataset GSA-FM, the sample sizes of four classes are 8, 15, 15, and 20 respectively, so the number of test samples for each class in 20 times of fivefold cross-validation is 200, 300, 300, and 400, respectively. And the number of correctly predicted samples are 200, 296, 295 and 290, respectively. The color of the diagonal elements in the confusion matrices is much heavier than the color of the nondiagonal elements, which shows that FE-ELM has high classification accuracy for the test samples of the two datasets, and the probability of the predicted class being equal to the real class is greater than 91.00% and 96.75%, respectively.

E. Compared to SVM on Dimensionality-Reduced Features

We also compare the proposed FE-ELM on the original features with SVM-K on dimensionality-reduced features obtained by different feature extraction methods. We perform the following classical feature extraction process on original GSA datasets. The maximum response value minus the baseline (MMB) is extracted from the time-series curves of the sensor array response [10]. The cubic polynomial coefficients (CPC) are extracted by fitting the time-series curves. The discrete Fourier transform coefficients (DFT) are extracted from the transform domain [32]. The relative difference maximum (RDM) is extracted based on the reconstructed phase space [33]. In addition, adaptive learning-based features such as linear discriminant analysis (LDA) [34] and autoencoder network (AEN) [35] are extracted as representative features.

Table V presents their classification accuracies, 95% confidence intervals, and time costs. It can be seen that FE-ELM achieves the best classification

accuracy. SVM-K has poor classification accuracy on dimensionality-reduced features due to the fact that many useful features are lost in the dimensionality-reduction process. While FE-ELM utilizes almost all the original features by integrating all base ELM, thus it achieves the superiority of classification. This suggests that the proposed method can be directly used for original GSA data classification, which is convenient compared to the traditional methods of dimensionality reduction first and then classification.

TABLE V Comparison of the Average Classification Results of FE-ELM on the Original Data With Those of SVK-K on the Dimensionality-Reduced Features by 20 Times Fivefold Cross-Validation

SECTION V.

Conclusion

In this article, we proposed an FE-ELM framework for GSA data classification. Unlike traditional methods that classify after feature dimensionality reduction, the proposed FE-ELM can be used directly on GSA data. The base classifiers of FE-ELM are trained on the fused feature subsets obtained by downsampling and sensor feature fusion, which not only overcomes the overfitting problem in GSA data classification, but also enables FE-ELM to achieve efficient and robust classification performance through ensemble learning. Experimental results show that the proposed FE-ELM outperforms traditional methods not only on a normal public dataset, but also on the dataset from low-concentration gases and is useful in extending the limit of detection of sensors. In the future, we will further develop the proposed approach on sensor array response data for more complex VOCs.